

Lecture 4

Evidence on Consumption Behavior

Bank of Portugal—Empirical Macroeconomics

Jonathon Hazell
London School of Economics, Autumn 2025

Overview

- Refresher—basic theory of consumption + testable implications
 - Random walk / permanent income hypothesis
 - Liquidity constraints
 - Precautionary saving
- Empirics of consumption
 - Macro data: tests of the random walk hypothesis
 - Micro data:
 - MPC estimation
 - Structural estimation of precautionary saving

Random-Walk Hypothesis

- Consumption Euler equation with uncertainty:

$$U'(C_t) = \beta(1+r)E_t[U'(C_{t+1})]$$

with flow budget constraint

$$A_{t+1} = (1+r)(A_t + Y_t - C_t)$$

- With $\beta(1+r) = 1$:

$$U'(C_t) = E_t[U'(C_{t+1})]$$

- Marginal utility is a martingale:

- Best current predictor of tomorrow's marginal utility is today's marginal utility

Random-Walk Hypothesis

- Suppose marginal utility is linear (i.e., quadratic utility):

$$C_t = E_t C_{t+1}$$

- Consumption a martingale!! Certainty equivalence:
 - C_t depends only on $E_t C_{t+1}$ not $\text{var}_t(C_{t+1})$ (or any higher moments)

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Very extreme model:

- Savings behavior unaffected by uncertainty!!
- Only consumption smoothing + intertemporal substitution affects savings

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Same thing for linearized or log-linearized models

- E.g. log-linearized New Keynesian model

Random Walk Hypothesis

- Random walk hypothesis (Hall 1978)

$$E_t C_{t+1} = C_t$$

$$C_{t+1} = C_t + \varepsilon_{t+1}$$

where $\varepsilon_{t+1} = C_{t+1} - E_t C_{t+1}$

- Two important properties of ε_{t+1} :
 - Since it is an expectations error, it is uncorrelated with information known at time t or earlier
 - It is proportional to the innovation to the consumer's present value of life-time income ("permanent income" for short)

Random Walk Hypothesis

In infinite horizon case:

$$C_t = \frac{r}{1+r} \left(A_t + \sum_{j=0}^{\infty} (1+r)^{-j} E_t Y_{t+j} \right)$$

$$\Delta C_{t+1} = \varepsilon_{t+1} = \frac{r}{1+r} \sum_{j=0}^{\infty} (1+r)^{-j} \Delta E_{t+1} Y_{t+1+j}$$

where $\Delta E_{t+1} Y_{t+1+j} = E_{t+1} Y_{t+1+j} - E_t Y_{t+1+j}$

Empirical Implications of the Random Walk Hypothesis

- Response to anticipated changes in income: Zero
- Response to transitory unanticipated change in income: Small
 - Marginal propensity to consume something like 2-5% per year
- Response to permanent unanticipated change in income: Large
 - 1% permanent increase in income raises “permanent income” by 1% (ignoring assets) and should therefore raise consumption by 1%

Overview

- Refresher—basic theory of consumption + testable implications
 - Random walk / permanent income hypothesis
 - **Liquidity constraints**
 - Precautionary saving

Liquidity Constraints

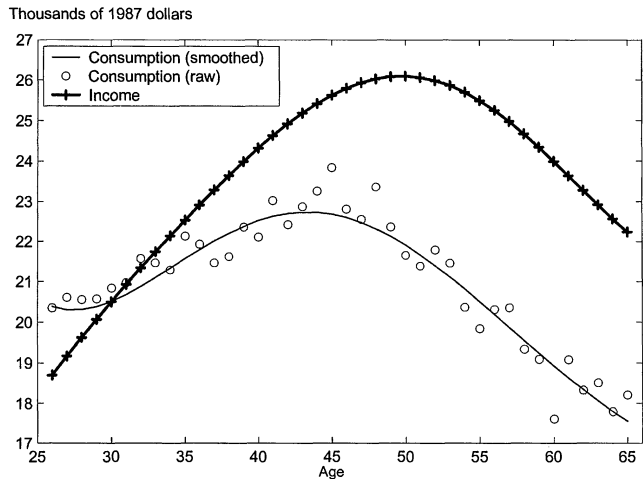


FIGURE 2.—Household consumption and income over the life cycle.

Source: Gourinchas-Parker (2002). Takes out cohort and time effects. Family size held constant over life-cycle.

Liquidity Constraints

- Consumption smoothing over the life-cycle likely involves substantial borrowing early in life
 - Simple PIH/LCH model assumes people can borrow (unsecured) at same rate as they can save
- Highly unrealistic:
 - Most household borrowing is secured (e.g., mortgages, car loans)
 - Interest rates on car loans and even mortgages substantially higher than on savings accounts
 - Interest rates on unsecured consumer lending (i.e., credit cards) extremely high ($\sim 20\%$)
 - Limits on unsecured borrowing beyond which can't go at any rate

Liquidity Constraints

Effects of liquidity constraints:

1. Less borrowing when they bind
2. Less borrowing even when they don't bind because they may bind in the future
 - Bad shock tomorrow may cause low consumption due to binding liquidity constraint at that point
 - Consumer saves today to “self-insure” against this future bad shock
3. Increases in income cause increases in consumption
 - Even if income shock is anticipated or transitory
 - If consumer is near the borrowing constraint

Buffer Stock Saving

- Liquidity constraints cause households facing uninsurable income risk to engage in **buffer stock saving** (i.e., self-insurance)
- Other sources of saving:
 - **Life-cycle saving** to smooth consumption over the life-cycle relative to life-cycle profile of income
 - Saving due to **patience/impatience**. If $1/\beta \neq (1+r)$ household will tilt consumption profile (down if impatient, up if patient)

Zeldes-Deaton-Carroll Model

- Households maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\gamma}}{1-\gamma},$$

subject to

$$W_{t+1} = R(W_t + Y_t - C_t) \quad W_t \geq 0$$

$$Y_t = P_t V_t$$

$$P_t = P_{t-1} N_t$$

where V_t and N_t are i.i.d. log-normal random variables.

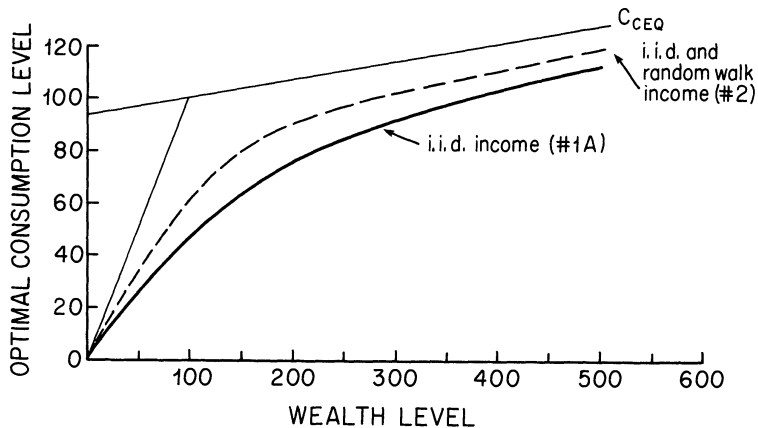
- R is given exogenously (partial equilibrium)
- Household income shocks are uninsurable

See Zeldes (1989), Deaton (1991), Carroll (1992, 1997)

Zeldes-Deaton-Carroll Model

- Zeldes-Deaton-Carroll predicts:
 1. Lower level of average consumption (buffer stock saving)
 2. High MPC out of transitory income shocks
 3. That consumption tracks income over the life-cycle

- Sometimes called the “buffer stock model”
 - General equilibrium version is Bewley-Aiyagari-Huggett model (i.e., interest rate is endogenous)



$T=15$

Income Processes: # 1A, # 2

$$U = \frac{C^{1-A}}{1-A} \quad A=3 \quad \text{Expected future income equals 100 per period}$$

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 - **Precautionary saving**

Precautionary Savings

- Curvature of utility almost surely falls as consumption rises:

$$U'''(C_t) > 0$$

(unlike quadratic utility as assumed for the random walk hypothesis)

- What does this imply about savings? With $U'''(C_t) > 0$, $U'(C_t)$ is convex
- If $C_t = E_t C_{t+1}$, then
 - $U'(C_t) < E_t U'(C_{t+1})$ (since $U'(E_t C_{t+1}) < E_t U'(C_{t+1})$)
 - Marginal reduction in C_t (increase in saving) increases expected utility
- Extra saving relative to certainty equivalence is **precautionary saving**

Precautionary Savings Similar To Liquidity Constraints

- Precautionary saving makes agents behave similarly to liquidity constraints
- I.e. greater income risk or more convex marginal utility similar effect to binding borrowing constraint

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- **Intuition #1:** income risk leads to “self imposed borrowing constraints”
 - Example: imagine Markovian labor income process, zero with positive probability
 - Positive probability of zero income forever
 - If agents borrow now they risk zero consumption in the future (which is very painful)
 - Zero is the “natural borrowing limit” (Requires Inada condition $\lim_{c \rightarrow 0} u'(c) = \infty$)

Precautionary Savings Similar To Liquidity Constraints

- **Intuition #2:** income risk leads to predictable consumption growth (assuming $\beta(1+r) = 1$)

$$u'(c_t) = E_t[u'(c_{t+1})]$$

second order Taylor expansion (preserves role for risk)

$$u'(c_t) = u'(c_t) + u''(c_t) E_t[c_{t+1} - c_t] + \frac{u'''(c_t)}{2} E_t[(c_{t+1} - c_t)^2]$$
$$\implies E_t\left[\frac{c_{t+1} - c_t}{c_t}\right] = \frac{1}{2} \psi_t E_t\left[\left(\frac{c_{t+1} - c_t}{c_t}\right)^2\right] \quad \psi_t \equiv \frac{u'''(c_t) c_t}{u''(c_t)}$$

Precautionary Savings Similar To Liquidity Constraints

Formal result (Carroll, Holm & Kimball 2021)

“We provide the analytical explanation of the interactions between precautionary saving and liquidity constraints. The effects of liquidity constraints and risks are similar because both stem from the same source: a concavification of the consumption function. Since a more concave consumption function exhibits heightened prudence, both constraints and risks strengthen the precautionary saving motive.”

From Basic Theory to Empirics

- Random walk hypothesis—**testable implications:**
 - Consumption is a random walk
 - MPC out of transitory income changes is zero

- Precautionary saving + liquidity constraints—**testable implications:**
 - Consumption far from a random walk, tracks income over life cycle
 - MPC out of transitory income shock > 0

Empirics of Consumption

Empirics of Consumption

- Macro data: tests of the random walk hypothesis
- Micro data:
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Early Tests

- Random walk hypothesis implies that consumption growth should be unpredictable. I.e., unforecastable using lagged variables
- Suggests following regression test:

$$\Delta C_{t+1} = \alpha + X_t\beta + \varepsilon_{t+1}$$

where X_t is a set of regressors known at time t and the test is $\beta = 0$

- Hall (1978) performed tests along these lines:
 - Failed to reject for lagged values of income and consumption
 - Rejected using lagged value of stock market
 - Interpreted results in favor of hypothesis

Campbell and Mankiw (1989)

- Rejections in early tests hard to interpret
 - Are they economically meaningful?
 - Even very accurate models can be rejected with enough data
- Useful to have a specific alternative hypothesis
- Two types of consumers:
 - Fraction λ are “hand-to-mouth”, i.e., consumer their income
 - Fraction $1 - \lambda$ are rational PIH consumers

Campbell and Mankiw (1989)

- Implies

$$\Delta C_t = \lambda \Delta Y_t + (1 - \lambda) \varepsilon_t$$

i.e., consumption growth is a weighted average of income growth

- Important complication:

- ΔY_t and ε_t are likely correlated. Why?

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- Implies

$$\Delta C_t = \lambda \Delta Y_t + (1 - \lambda) \varepsilon_t$$

i.e., consumption growth is a weighted average of income growth

- Important complication:

- ΔY_t and ε_t are likely correlated. Why?
- Recall that ε_t denotes innovations to permanent income
- Changes in current income likely correlated with innovations to permanent income

An IV Approach

$$\Delta C_t = \lambda \Delta Y_t + (1 - \lambda) \varepsilon_t$$

- Can we think of instruments that will work in this case? (Hint: Error term is an expectation error)

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$$\Delta C_t = \lambda \Delta Y_t + (1 - \lambda) \varepsilon_t$$

- Can we think of instruments that will work in this case? (Hint: Error term is an expectation error)
Any variable known at time $t - 1$ works as an instrument
- Since ε_t is an expectation error, it is orthogonal to all variables known at time $t - 1$ or earlier
- So, we can use lags of anything as instruments (Wow, lots of possible instruments)

Campbell & Mankiw—Main Takeaways

- Estimate λ of roughly 0.5 (half of consumers are “hand to mouth”)
- Strongly reject $\lambda = 0$ (random walk hypothesis)
- Lagged income growth weak instruments
- Lagged consumption growth much stronger instruments
 - Consumption seems to encode information about future income growth
- This type of rejection of random walk hypothesis is often referred to as “excess sensitivity”

Limitations of Macro-Data Tests

- Few observations
- Difficult to find variables with much predictive power for income
 - Weak instruments are a problem
- Rely on strong assumption that ε_t is only a expectations error
 - If not true, hard to find an exogenous instrument
 - Exercise: show that if $U = U(C_t, Z_t)$ where Z_t is a predictable shifter of utility, then valid instrument must be orthogonal to Z_{t+1}

Empirics of Consumption

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Empirics of Consumption

- Macro data: tests of the random walk hypothesis
- **Micro data:**
 - **MPC estimation**
 - Structural estimation of precautionary saving
- The macro data isn't informative so let's go to micro data!

Tests Using Micro Data

Large literature has analyzed anticipated changes in household level income:

- Parker (1999): Reaching Social Security payroll cap
- Souleles (1999): Receipt of tax refund
- Johnson, Parker, Souleles (2006): 2001 tax rebate
- Parker, Souleles, Johnson, McClelland (2013): 2008 tax rebate
- Hsieh (2003) and Kueng (2015): Alaska Permanent Fund payments
- Baker (2015): shocks to firm wide earnings
- Many papers: Covid stimulus payments ...

Anticipated or Unanticipated

- All these income changes are pre-announced
- But many were (likely) not very salient to households
 - I received 2020 rebate in the mail and was pleasantly surprised
- Does it matter whether consumers knew?

Anticipated or Unanticipated

- All these income changes are pre-announced
- But many were (likely) not very salient to households
 - I received 2020 rebate in the mail and was pleasantly surprised
- Does it matter whether consumers knew?
- If transitory, probably not that much
 - Don't affect permanent income much
 - MPC out of transitory income shock should be very small according to random walk hypothesis
- If persistent (e.g. Baker 2015) matters more

Parker-Souleles-Johnson-McClelland 2013

- Economic Stimulus Act of February 2008
- \$100 billion of tax rebates to 130 million US tax filers
 - Single filers received \$300-\$600 (max of \$300 and tax liability up to \$600)
 - Couples received \$600-\$1200
 - Phased out for incomes above \$75,000 (\$150,000 for couples)
- Timing of disbursement based on last two digits of SSN (effectively random)
- Compare spending of households that received payment at different dates
- **A natural experiment!**

TABLE 1—THE TIMING OF THE ECONOMIC STIMULUS PAYMENTS OF 2008

<i>Payments by electronic funds transfer</i>		<i>Payments by mailed check</i>	
Last two digits of taxpayer SSN	Date ESP funds transferred to account by	Last two digits of taxpayer SSN	Date check to be received by
00–20	May 2	00–09	May 16
21–75	May 9	10–18	May 23
76–99	May 16	19–25	May 30
		26–38	June 6
		39–51	June 13
		52–63	June 20
		64–75	June 27
		76–87	July 4
		88–99	July 11

Source: Internal Revenue Service (<http://www.irs.gov/newsroom/article/0,,id=180247,00.html>).

Nature of Income Shock

- Should be anticipated (program highly publicized)
 - Consumption response should be a lower bound on response of an unanticipated income change
 - Some of the effect may have occurred upon announcement
- Totally transitory
- **An example of the value of identified moments!**

Data

- Main data source: Consumer Expenditure Survey
- Households surveyed 4 times with 3 month intervals about spending over past 3 months
- New households added each month Authors worked with BLS to add questions about receipt of stimulus payments
 - Did they receive stimulus payment?
 - When did they receive it?
 - How much did they receive?
- **An example of creative empirical work!**

Empirical Specification

$$C_{i,t+1} - C_{i,t} = \sum_s \beta_{0s} \times month_{s,i} + \beta'_1 X_{i,t} + \beta_2 ESP_{i,t+1} + u_{i,t+1}$$

- Dependent variable: 3-month change in consumption
- Independent variable of interest: $ESP_{i,t+1}$
- Time dummies:
 - Soaks up all aggregate effects (GE effects, anticipation effects)
 - Identification comes from cross section
 - Comparison of those that get ESP at time $t+1$ and those that don't
- $X_{i,t}$ to soak up some variation from error term
- Are there threats to identification?

TABLE 2—THE CONTEMPORANEOUS RESPONSE OF EXPENDITURES TO ESP RECEIPT AMONG ALL HOUSEHOLDS

	Food OLS	Strictly nondurables OLS	Nondurable spending OLS	All CE goods and services OLS	Food OLS	Strictly nondurables OLS	Nondurable spending OLS	All CE goods and services OLS
<i>Panel A. Dollar change in spending</i>								
<i>ESP</i>	0.016 (0.027)	0.079 (0.046)	0.121 (0.055)	0.516 (0.179)				
<i>I(ESP)</i>					10.9 (31.7)	74.8 (56.6)	121.5 (67.2)	494.5 (207.2)
	Food OLS	Strictly nondurables OLS	Nondurable spending OLS	All CE goods and services OLS	Food 2SLS	Strictly nondurables 2SLS	Nondurable spending 2SLS	All CE goods and services 2SLS
<i>Panel B. Percent change in spending</i>					<i>Panel C. Dollar change in spending</i>			
<i>ESP</i>					0.012 (0.033)	0.079 (0.060)	0.128 (0.071)	0.523 (0.219)
<i>I(ESP)</i>	0.69 (1.27)	1.74 (0.96)	2.09 (0.94)	3.24 (1.17)				

Notes: All regressions also include a full set of month dummies, age, change in the number of adults, and change in the number of children following equation (1). Reported standard errors are adjusted for arbitrary within-household correlations and heteroskedasticity. The coefficients in panel B are multiplied by 100 so as to report a percent change. The last four columns report results from 2SLS regressions where the indicator variable for ESP receipt and the other regressors are used as instruments for the amount of the ESP. All regressions use 17,478 observations except for the first two columns of panel B which have only 17,427 and 17,475, respectively.

Types of Variation: First Pass

- Timing of payments is random
- How much a person got is not random
 - Correlated with income
 - Possible that this is correlated with error term (if high income people did relatively well or badly in this period)
- 2nd set of results use $I(ESP > 0)$ (i.e., only whether household received ESP, not how much)

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- 2nd set of results use $I(ESP > 0)$ (i.e., only whether household received ESP, not how much)
- Panel C uses 2SLS with $I(ESP > 0)$ as an instrument for $ESP_{i,t+1}$
 - First stage: $ESP_{i,t+1}$ on $I(ESP > 0)$
 - Reduced form: $C_{i,t+1} - C_{i,t}$ on $I(ESP > 0)$
 - IV is ratio of these two

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Types of Variation: 2nd Pass

- Timing of payments is random
- Who got payments is not random (again, correlated with income)

Types of Variation: 2nd Pass

- Timing of payments is random
- Who got payments is not random (again, correlated with income)
- Three approaches:
 - Control for receipt of payment
 - Only households that received payment
 - Only households that reported receiving payment on time
- Most of later results with this last sample
 - Results significant, but standard errors not trivial

TABLE 3—THE RESPONSE TO ESP RECEIPT AMONG HOUSEHOLDS RECEIVING PAYMENTS

	Dollar change in		Percent change in		Dollar change in	
	Nondurable spending OLS	All CE goods and services OLS	Nondurable spending OLS	All CE goods and services OLS	Nondurable spending 2SLS	All CE goods and services 2SLS
<i>Panel A. Sample of all households (N = 17,478)</i>						
<i>ESP</i>	0.117 (0.060)	0.507 (0.196)			0.123 (0.081)	0.509 (0.253)
<i>I(ESP)</i>			2.63 (1.07)	3.97 (1.34)		
<i>I(ESP_{i,t} > 0 for any t)_i</i>	9.58 (36.07)	21.21 (104.00)	−0.88 (0.50)	−1.17 (0.63)	8.23 (38.79)	20.77 (112.18)
<i>Panel B. Sample of households receiving ESPs (N = 11,239)</i>						
<i>ESP</i>	0.185 (0.066)	0.683 (0.219)			0.252 (0.103)	0.866 (0.329)
<i>I(ESP)</i>			3.91 (1.33)	5.63 (1.69)		
<i>Panel C. Sample of households receiving only on-time ESPs (N = 10,488)</i>						
<i>ESP</i>	0.214 (0.070)	0.590 (0.217)			0.308 (0.112)	0.911 (0.342)
<i>I(ESP)</i>			4.52 (1.50)	6.05 (1.89)		

Source: Parker et al. (2013)

Longer-Term Impact

- Do effects reverse or build over time?
- Add lagged term to regression

TABLE 5—THE LONGER-RUN RESPONSE OF EXPENDITURES TO ESP RECEIPT

	Dollar change in		Percent change in		Dollar change in	
	Nondurable spending OLS	All CE goods and services OLS	Nondurable spending OLS	All CE goods and services OLS	Nondurable spending 2SLS	All CE goods and services 2SLS
ESP_{t+1} or $I(ESP_{t+1})$	0.201 (0.067)	0.517 (0.211)	3.92 (1.55)	4.96 (1.96)	0.254 (0.110)	0.757 (0.360)
ESP_t or $I(ESP_t)$	-0.054 (0.080)	-0.288 (0.214)	-1.23 (1.50)	-2.22 (1.92)	-0.097 (0.113)	-0.278 (0.330)
Implied spending effect in second three-month period	0.146 (0.104)	0.230 (0.303)	NA	NA	0.156 (0.177)	0.479 (0.568)
Implied cumulative fraction of rebate spent over both three-month periods	0.347 (0.155)	0.747 (0.477)	NA	NA	0.410 (0.273)	1.235 (0.892)

Notes: All regressions also include the change in the number of adults, the change in the number of children, the age of the household, and a full set of month dummies. The sample includes only households receiving only on-time ESPs. Standard errors are adjusted for arbitrary within-household correlations and heteroskedasticity. The coefficients in the second triplet of columns are multiplied by 100 so as to report a percent change. The final triplet of columns reports results from 2SLS regressions where $I(ESP)$ and the other regressors are used as instruments for ESP . The number of observations for all regressions is 10,488.

Longer-Term Impact

- Do effects reverse or build over time?
- Add lagged term to regression
 - Growth slightly negative in next quarter
 - But level still above control group
- Point estimate thus suggests higher spending persists and does not reverse
- Standard errors large

What Did They Spend It On?

- Compare share of increase with share of spending

TABLE 7—THE PROPENSITY TO SPEND ON SUBCATEGORIES OF EXPENDITURES

Dependent variable:	<i>Panel A. Food</i>			<i>Panel B. Additional categories in strictly nondurables</i>			
	Food at home	Food away from home	Alcoholic beverages	Utilities, household operations	Personal care and misc.	Gas, motor fuel, public transportation	Tobacco products
Coefficient on ESP	0.050	0.025	0.011	0.059	0.083	0.027	0.007
Standard error	(0.032)	(0.033)	(0.007)	(0.027)	(0.049)	(0.039)	(0.009)
Implied share of increase in nondurable spending	0.16	0.08	0.04	0.19	0.27	0.09	0.02
Share of avg. spending on subcategory	0.23	0.11	0.01	0.23	0.04	0.16	0.01
Dollar change in spending on:	<i>Panel C. Additional categories in nondurables</i>			<i>Panel D. Additional categories in total CE spending</i>			
	Apparel	Health	Reading	Housing (incl. furnishings)	Entertainment	Education	Transportation
Coefficient on ESP	0.022	0.025	−0.001	0.099	0.077	−0.100	0.527
Standard error	(0.021)	(0.048)	(0.003)	(0.092)	(0.099)	(0.042)	(0.269)
Implied share of increase in:							
Nondurable spending	0.07	0.08	0.00				
Durable spending				0.16	0.13	−0.17	0.87
Avg. spending on subcategory:							
Share of nondurable	0.06	0.15	0.01				
Share of durable				0.56	0.13	0.04	0.27

Source: Parker et al. (2013)

Panel E. Subcategories of transportation

Dollar change in spending on:	New vehicle purchases	Used vehicle purchases	Other vehicle purchases	Maintenance and repairs	Other, insurance fees, etc.
Coefficient on ESP	0.357	0.123	0.011	0.009	0.027
Standard error	(0.204)	(0.149)	(0.054)	(0.028)	(0.024)
Implied share of increase in durable spending	0.59	0.20	0.02	0.01	0.04
Share of average durable spending	0.07	0.06	0.01	0.04	0.09

Source: Parker et al. (2013)

What Did They Spend It On?

- Compare share of increase with share of spending
- For non-durables: alcohol, personal care, tobacco, apparel
- For durables: cars
- Large effect on cars suggests ESP provided down payment for debt-financed cars (alleviated liquidity constraints)
 - Possible reversal for cars:
 - Did it move everyone forward a few months (no reversal)?
 - Or did those that didn't buy immediately, spend it on something else and become liquidity constrained again (subsequent reversal)?
 - Some evidence of reversal in other contexts ("cash for clunkers", Mian & Sufi 2012)

Durable versus Non-Durable Consumption

- What is the most relevant statistic?
 - MPC of non-durables?
 - MP of total spending?

Durable versus Non-Durable Consumption

- What is the most relevant statistic?
 - MPC of non-durables?
 - MP of total spending?
- Depends on the question and model!
 - Non-durable consumption corresponds more closely to utility flow of consumption in canonical model
 - MP of total spending affects “bang for buck” of policy
See (Laibson, Maxted & Moll 2022)

Criticism of Parker et al (2013)

1. Imprecision
2. “Forbidden comparisons” and the practice of event studies

Imprecision

TABLE 3—THE RESPONSE TO ESP RECEIPT AMONG HOUSEHOLDS RECEIVING PAYMENTS

	Dollar change in		Percent change in		Dollar change in	
	Nondurable	All CE goods	Nondurable	All CE goods	Nondurable	All CE goods
	spending	and services	spending	and services	spending	and services
	OLS	OLS	OLS	OLS	2SLS	2SLS
<i>Panel A. Sample of all households (N = 17,478)</i>						
<i>ESP</i>	0.117 (0.060)	0.507 (0.196)			0.123 (0.081)	0.509 (0.253)
<i>I(ESP)</i>			2.63 (1.07)	3.97 (1.34)		
<i>I(ESP_{i,t} > 0 for any t)_i</i>	9.58 (36.07)	21.21 (104.00)	−0.88 (0.50)	−1.17 (0.63)	8.23 (38.79)	20.77 (112.18)
<i>Panel B. Sample of households receiving ESPs (N = 11,239)</i>						
<i>ESP</i>	0.185 (0.066)	0.683 (0.219)			0.252 (0.103)	0.866 (0.329)
<i>I(ESP)</i>			3.91 (1.33)	5.63 (1.69)		
<i>Panel C. Sample of households receiving only on-time ESPs (N = 10,488)</i>						
<i>ESP</i>	0.214 (0.070)	0.590 (0.217)			0.308 (0.112)	0.911 (0.342)
<i>I(ESP)</i>			4.52 (1.50)	6.05 (1.89)		

Forbidden Comparisons and Event Studies

- Recall the specification:

$$C_{i,t+1} - C_{i,t} = \sum_s \beta_{0s} \times month_{s,i} + \beta_1' X_{i,t} + \beta_2 ESP_{i,t+1} + u_{i,t+1}$$

- Assuming that the rebate is randomly assigned:
 - *What is the treatment group and what is the control group in Parker et al?*

Forbidden Comparisons and Event Studies

- Recall the specification:

$$C_{i,t+1} - C_{i,t} = \sum_s \beta_{0s} \times month_{s,i} + \beta_1' X_{i,t} + \beta_2 ESP_{i,t+1} + u_{i,t+1}$$

- Assuming that the rebate is randomly assigned:
 - *What is the treatment group and what is the control group in Parker et al?*
- Treatment group:
 - People who received the rebate this quarter
- Control group:
 - People who received rebate **last quarter or next quarter**
 - Part of the control group were **previously treated!**
 - Not obvious that Parker et al estimates the MPC out of transitory income shocks

Forbidden Comparisons

- General problem of **forbidden comparisons**
 - Control group in an event study has previously been treated
 - Leads to bias except under implausible treatment effect homogeneity
 - See Borusyak, Jaravel & Spiess 2022 for a general explanation of the problem
 - Parker et al regression is a specific example
- Numerical example

Forbidden Comparisons—Bias

Borusyak, Jaravel & Spiess 2022:

“Our findings deliver several insights for the macroeconomics literature. While commonly used estimates of the quarterly MPX covering all expenditures range from 50-90% and estimates of the quarterly MPX for nondurable expenditure range from 15-25%,¹ our estimates, when appropriately rescaled, are about half as large, at 25–37% for the MPX one quarter after tax rebate receipt for all expenditures and 8–11% for nondurables.”

Forbidden Comparisons—Solutions

■ Solutions:

1. **Reduced form correction.** Apply a regression technique that avoids forbidden comparisons.
 - Many to choose from, but the best is by Borusyak, Hull & Jaravel (2022)
 - Borusyak, Hull & Jaravel (2022) and Ramey, Orchard & Wieland (2023) apply this technique to Parker et al
2. **Structural correction and simulated method of moments.** (cf. Kaplan & Violante 2014)
 - Obtain “the true” structural model
 - Randomly give consumers in the model a tax rebate and simulate data
 - Run the Parker et al on the simulated data
 - Choose structural parameters (e.g. discount factor, coefficient of risk aversion) so that regression coefficients are same in simulated and real-world data

Parker et al.—Takeaways

- Landmark study with creative natural experiment suggests MPC is large
- Follow up work:
 - Data is noisy
 - Regression is mis-specified
 - MPC still positive but smaller than naive interpretation

Hsieh (2003)

- Parker (1999) and Souleles (1999) find that households respond to predictable changes in income
 - Parker (1999): Households hitting social security tax limit
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 1. Failure of “LC-PIH”
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- Browning and Collado (2001) study large predictable seasonal variation in earnings in Spain and find no response of consumption

Hsieh (2003)

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- Payments are large and predictable

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- Concludes: Households will behave according to "LC-PIH" when it comes to large and regular payments
(It turns out this conclusion was wrong)

The Alaska Permanent Fund

- Created in 1976
- 25% of states government's oil royalties go to fund
- Since 1982, about 50% of fund dividends distributed to Alaskan residents
- Subject to eligibility, every resident gets the same amount
 - Amount equal to $\text{payment}_t \times \text{family size}_h$
 - Over sample period, payments varied from low of \$331 in 1984 to high of \$1,964 in 2000
- Good for testing "LC-PIH":
 - Payments are large and predictable
 - Application in March. Disbursement in October. Amount set in September. But estimated by newspapers before that.

Data

- Main data source: Consumer expenditure survey
- Aggregates observations to household level
- Drops households in student housing, lacking family size, age of head of household, or food expenditures. Also drops movers.
- Total number of observations: about 800

Excess Sensitivity Test

$$\log \left(\frac{C_{th}^{IV}}{C_{th}^{III}} \right) = \alpha_1 \frac{PFD_t \times \text{family size}_h}{\text{family income}_h} + z'_{th} \alpha_2 + \varepsilon_{th}$$

- C_{th}^{IV} is non-durable consumption of household h in quarter IV
- PFD_t is Permanent Fund payout per person in year t
- z_h contains constant, change in # adults, # children, 2nd order polynomial in age of household head
- α_1 measures elasticity of household consumption with respect to increase in income due to Permanent Fund payments

TABLE 2—RESPONSE OF CONSUMPTION TO ALASKA PFD

	dlog(<i>Nondurable consumption</i>)			dlog(<i>Durable consumption</i>)		
	(1)	(2)	(3)	(4)	(5)	(6)
$PFD_t \times Family\ Size_h$	0.0002	-0.0167	-0.0034	-0.1659	-0.1741	-0.1488
$Family\ Income_h$	(0.0324)	(0.0336)	(0.0328)	(0.0878)	(0.0916)	(0.0890)
<i>Controls for:</i>						
<i>Family size</i>	No	No	Yes	No	No	Yes
<i>Year dummies</i>	No	Yes	No	No	Yes	No
Number of observations	806	806	806	806	806	806

Notes: Dependent variable is $\log(C_{IV}/C_{III})$. Standard errors are in parentheses. All regressions are ordinary least squares (OLS) and include a quadratic in age and changes in the number of children and adults in the household.

No Excess Sensitivity for Non-Durables

- Baseline elasticity for non-durable consumption 0.0002 (s.e. 0.0324)
 - Typical shock 20%. Response 0.004 percent or 4 cents.
 - Baseline estimated from variation across years and across family size

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- Baseline elasticity for non-durable consumption 0.0002 (s.e. 0.0324)
 - Typical shock 20%. Response 0.004 percent or 4 cents.
 - Baseline estimated from variation across years and across family size
- Perhaps seasonal pattern is different for households of different size
- 3rd column controls for family size (only uses variation across time)

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Negative Sensitivity for Durables

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- Suggests households purchase durables in 3rd quarter, before payments are made
- This is consistent with theory, since this is when payment amount becomes known

Why Are Results So Different from Previous Literature?

- Perhaps Alaskan households less liquidity constrained
- But they are substantially younger ...
- And results hold for those with low income
- Perhaps due to size and visibility of payments
 - Check this by considering response to income tax receipts (as in Souleles 1999)

TABLE 6—RESPONSE OF NONDURABLE CONSUMPTION TO
INCOME TAX REFUNDS AND PFD

	dlog(<i>Nondurable consumption</i>)	
	log(C_{II}/C_I)	log(C_{IV}/C_{III})
$PFD_t \times \text{Family Size}_h$	—	0.0032
Family Income_h		(0.0562)
$\text{Income tax refund}_h$	0.2831	—
Family Income_h	(0.1140)	
Number of observations	369	369

Notes: Dependent variable is $\log(C_{II}/C_I)$ in the first column and $\log(C_{IV}/C_{III})$ in the second column. Standard errors are in parentheses. All regressions are OLS and include a quadratic in age and changes in the number of children and adults in the household.

Hsieh (2003): Conclusion

- Households display excess sensitivity to small, unpredictable, hard to predict changes in income
 - Consistent with Parker (1999), Souleles (1999), Johnson-Parker-Souleles (2006), Parker et al. (2013)
- Households do not display excess sensitivity to large, predictable, highly visible changes in income
 - Consistent with Browning and Collado (2001)
- Combination of results consistent with “near rationality”

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- What could have gone wrong?

Kueng (2015)

- Revisits Hsieh's (2003) analysis and gets very different results

Kueng (2015)

- Revisits Hsieh's (2003) analysis and gets very different results
- Normalizes dividend payments by total expenditure as opposed to current total family pre-tax income
- This makes a big difference
- Lots of measurement error in family income variable
 - Attenuation bias
- Extends sample by 12 years and uses non-Alaskans as control group
 - Much more variation in dividend payments
 - Control group also improves precision

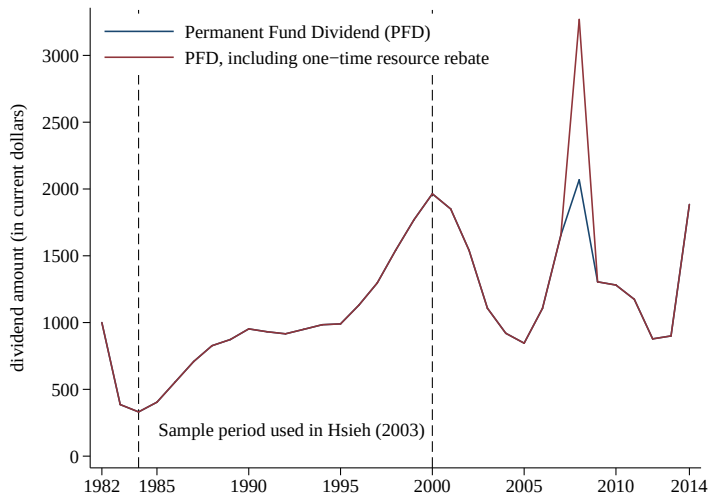


Figure 2 – Alaska Permanent Fund Dividend per person, 1982-2014 (nominal amount)

Specification

$$\log c_{i,t} - \log c_{i,t-1} = \alpha_1 \frac{\text{PFD}_t \times \text{family size}_i}{y_i} + \alpha_2' z_{i,t} + \varepsilon_{i,t}$$

- y_i is either total expenditures or pre-tax total income
- $z_{i,t}$ is a vector of controls that may include fixed effects

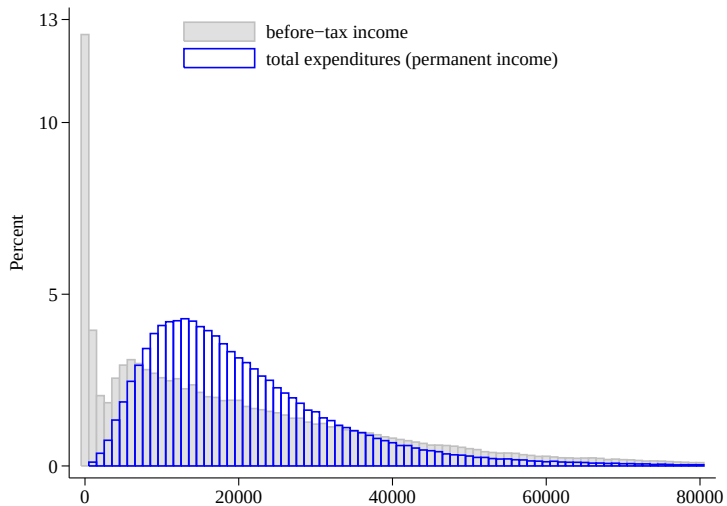


Figure 1 – Distribution of annual before-tax family income and total annualized expenditures

Table 2: Spending excess sensitivity tests using the Permanent Fund Dividend

Dep. var.: $\Delta \ln(c_{it})$, nondurables and services	<i>Alaskans only</i>					<i>All households</i>			
	Hsieh's specification		normalize w/ total expend.	control for aggr. effects	more sample selection	using rest of U.S. as control	control for all main effects	attenuation factor	IV curr inc w/ perm inc
	Hsieh (2003)	replication and extension							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<u>A: Sample 1980-2001</u>									
PFD x family size x Alaska / before-tax income	-0.003 (0.033)	-0.003 (0.005)							0.052** (0.025)
PFD x family size x Alaska / total expenditures			0.123 (0.086)	0.124 (0.112)	0.126 (0.127)	0.090** (0.036)	0.091** (0.036)	0.107** (0.043)	
Number of observations (rounded)	806	800	800	800	600	315200	315200	315200	281500
Number of Alaskan obs. (rounded)	806	800	800	800	600	4300	4300	4300	3800
Number of clusters (rounded)	—	0	800	800	600	117000	117000	117000	103400
Number of Alaskan CUs (rounded)	806	800	800	800	600	1700	1700	1700	1500
R-squared	N/A	0.009	0.013	0.038	0.044	0.009	0.009	0.009	0.010
F-statistic for current and lagged dividend									

Source: Kueng (2015)

Kueng (2015) – Main Results

- Normalizing by total expenditures dramatically changes results
- Results similar for extended sample (more significant without time FE)
- Using non-Alaskans as a control group improves precision (Why?)
- Column 8 takes into account that on average only 83 cents per dollar of PFD is received in the form of cash income (some is garnished, also college fund, etc.)

Kueng (2018)

- Kueng' (2018): “Why are MPCs so high even for large and pre-announced transfers”?
- Introduces a **new dataset**
 - Account level aggregator i.e. Mint.com (?)
 - Has data on consumption expenditures, demographics **and liquid balances**
 - Large (though short 3 year time series)
- Estimation:

$$\Delta c_{it} = \sum_s \beta_s \text{Permanent Fund Dividend}_{i,t-s} + \alpha_t + \text{Alaska}_i + \text{controls}_{\varepsilon_{it}}$$

- Δc_{it} is monthly spending changes, β_s is seasonal dummies, α_t is time fixed effects
- Sample is households in Alaska (1300 accounts) and Washington state (2200 accounts)
- What is the treatment group and what is the control group?

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Concerns with Treatment/Control Comparison?

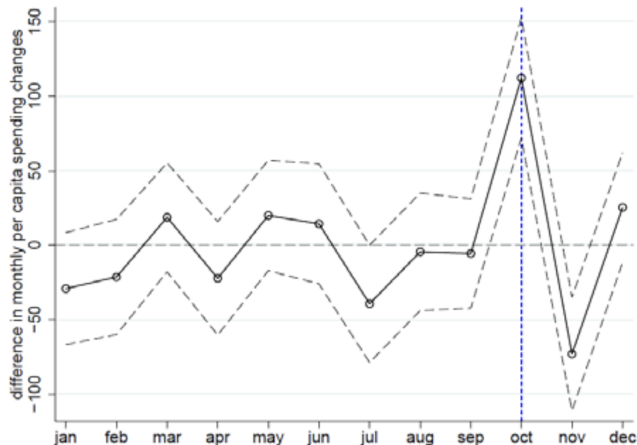
Concerns with Treatment/Control Comparison?



Treatment/Control Comparison and Main Result

- Consumption growth similar for treatment/control except during dividend
 - I.e. no pre-trends

Treatment/Control Comparison and Main Result



Notes: This figure shows the average difference in monthly household per capita spending changes of nondurables and services between households in Alaska and Washington. The Permanent Fund Dividend is paid out at the beginning of October. Per capita spending is calculated using the OECD household equivalence scale. Dashed lines are 95% confidence intervals.

Main Estimates

- Implied MPC from arrival: 1 Month: 12% 1 Quarter: 22%
 - Large MPCs!
- Consistent with the buffer stock model?
 - Large MPCs
 - No anticipation effects though dividend is salient

Inconsistencies with Buffer Stock Model

MPC Heterogeneity by Quintiles of Liquidity and Income

Interaction measure:	Panel A: Liquidity		Panel B: Income	
	Liquid assets	CoH ratio	Current income	Permanent income
	(1)	(2)	(3)	(4)
PFD $\times$ first quintile	0.270*** (0.065)	0.357*** (0.059)	0.117** (0.051)	0.080 (0.072)
PFD $\times$ second quintile	0.283*** (0.057)	0.253*** (0.065)	0.079 (0.068)	0.163*** (0.055)
PFD $\times$ third quintile	0.237*** (0.085)	0.292*** (0.101)	0.291*** (0.070)	0.163** (0.069)
PFD $\times$ fourth quintile	0.181* (0.106)	0.190* (0.098)	0.371*** (0.105)	0.304*** (0.092)
PFD $\times$ fifth quintile	0.341*** (0.093)	0.207** (0.095)	0.572*** (0.113)	0.761*** (0.116)
PFD				
$\left(\frac{\text{PFD}}{100}\right)^2$				
Observations	44,577	44,577	44,577	44,577
R-squared	0.128	0.128	0.129	0.130
$p(\beta_1 = \beta_5)$	0.5132	0.1557	0.0001	0.0000

Inconsistencies with Buffer Stock Model

- Low and high liquidity households have similar MPCs
- Rich households have higher MPCs:
 - Reason: payment is small relative to permanent income
- Rejects the canonical model

Explanations and Summary of Kueng (2018)

- Kueng's explanation is "near rationality"
 - Transfers are a smaller share of permanent income for rich people
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 - In response to large and pre-announced transfer
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 - What's the "right" model of consumption??
- This is a great paper! Due to a great data accessed during grad school

Empirics of Consumption

- Macro data: tests of the random walk hypothesis
- Micro data:
 - MPC estimation
 - **Structural estimation of precautionary saving**

Excess Sensitivity over the Life-Cycle

Up until now:

- Excess sensitivity of consumption to predictable movements in income

Another type of potential excess sensitivity:

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Use moment as input into structural estimation of consumption model

Life-Cycle Consumption Needs

- Consumption needs may vary over the life-cycle
 - Most obvious source of such variation is family size and composition
- Gourinchas and Parker (2002) study U.S. household consumption data
 - From the Consumer Expenditure Survey
 - Regress household consumption on fixed effects for time, birth cohort and family size
 - Then plot residual consumption
- Consumption still tracks income over the life cycle
 - Suggests liquidity constraints / precautionary saving matters

Thousands of 1987 dollars

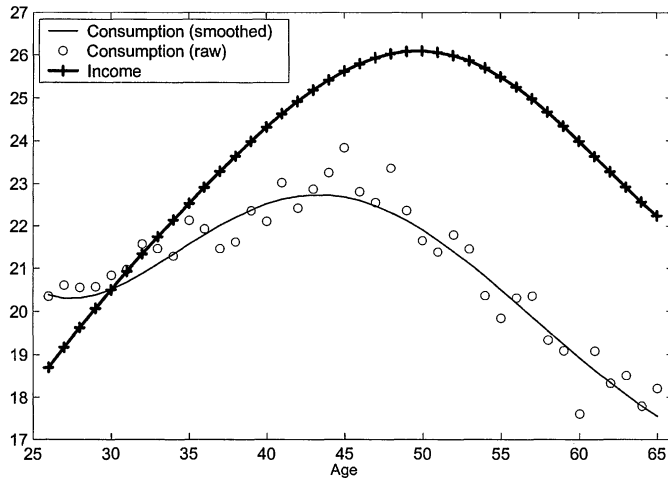


FIGURE 2.—Household consumption and income over the life cycle.

Source: Gourinchas-Parker (2002). Takes out cohort and time effects. Family size held constant over life-cycle.

Gourinchas-Parker 2002

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 - Estimate the parameters from Zeldes-Deaton-Carroll model
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- **Note:** consumption + income path are clearly not “identified moments”
 - But clearly useful for pinning down precautionary motive

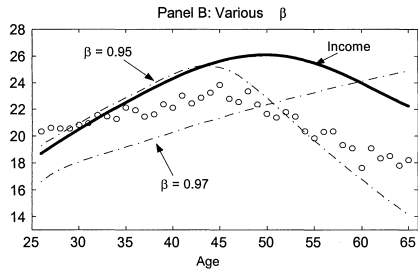
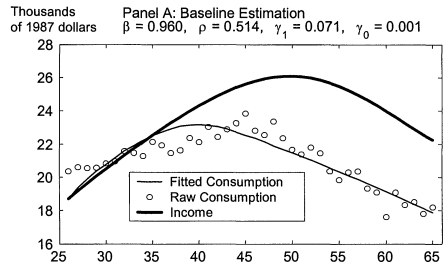
TABLE III
STRUCTURAL ESTIMATION RESULTS

MSM Estimation	Robust Weighting	Optimal Weighting
Discount Factor (β)	0.9598	0.9569
S.E.(A)	(0.0101)	
S.E.(B)	(0.0179)	(0.0150)
Discount Rate ($\beta^{-1} - 1$)(%)	4.188	4.507
S.E.(A)	(1.098)	
S.E.(B)	(1.949)	(1.641)
Risk Aversion (ρ)	0.5140	1.3969
S.E.(A)	(0.1690)	
S.E.(B)	(0.1707)	(0.1137)
Retirement Rule:		
γ_0	0.0015	$5.68 \cdot 10^{-6}$
S.E.(A)	(3.84)	
S.E.(B)	(3.85)	(16.49)
γ_1	0.0710	0.0613
S.E.(A)	(0.1215)	
S.E.(B)	(0.1244)	(0.0511)
$\chi^2(A)$	175.25	
$\chi^2(B)$	174.10	185.67

Note: MSM estimation for entire group. Standard errors calculated without (A) and with (B) correction for first stage estimation. Cell size is 36,691 households. The last row reports a test of the overidentifying restrictions distributed as a Chi-squared with 36 degrees of freedom. The critical value at 5% is 50.71. Efficient estimates are calculated with a weighting matrix $\hat{\Omega}$ computed from the robust estimates.

Results

- Reasonable discount rate of 4%
 - Reasonable IES of about 2 (sensitive to weighting matrix)
 - Reasonable MPC in retirement of 7% per year



Target Cash on Hand

- Level of cash on hand at age t that is expected to remain unchanged:

$$\bar{x}_t = E_t[x_{t+1} | x_t = \bar{x}_t]$$

- If $x_t > \bar{x}_t$ households dissave on average
- If $x_t < \bar{x}_t$ households build up assets on average

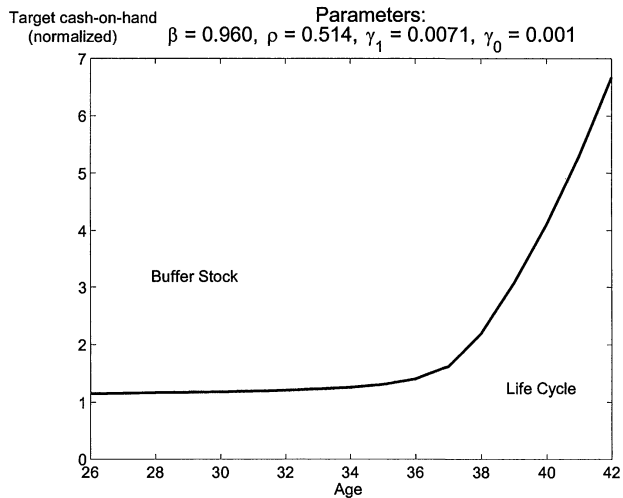


FIGURE 6.—Normalized target cash-on-hand by age.

Source: Gourinchas-Parker (2002). Cash-on-hand normalized by permanent income.

Life-Cycle Consumption Savings

- Define life-cycle consumption as consumption under complete insurance
 - I.e. a hypothetical scenario in the model
 - All households pool income risk
- Use this concept to construct:
 - Life-cycle savings / wealth
 - Buffer-stock savings / wealth

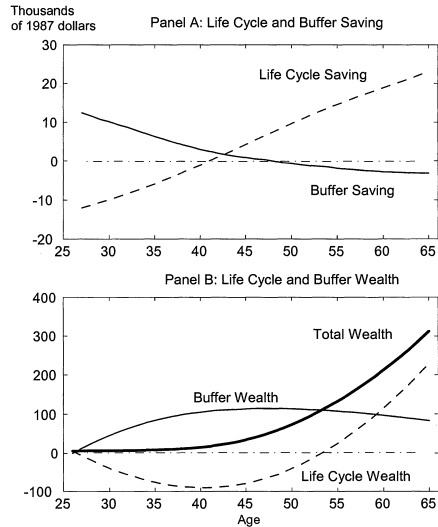


FIGURE 7.—The role of risk in saving and wealth accumulation.

Gourinchas-Parker 02: Summary

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Gourinchas-Parker 02: Summary

- Can uninsurable income risk explain comovement of consumption and income over the life-cycle?
- Yes!
 - Households are impatient (want downward sloping consumption profiles)
 - Consumption constrained by income early in life
 - Households save for retirement later in life
- Challenges:
 1. Can this model quantitatively match MPC out of transitory income shocks?
 2. Is the hump shape in income and consumption due to borrowing constraints?

Kaplan and Violante (2014)

- Gourinchas-Parker model can't match large estimated spending responses of consumption to tax rebates
 - *Qualitatively* Gourinchas-Parker model predicts positive MPCs out of transitory income
 - *Quantitatively* they are much too low
 - For reasonable calibrations of wealth, households “save out of borrowing constraint”

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 - *Qualitatively* Gourinchas-Parker model predicts positive MPCs out of transitory income
 - *Quantitatively* they are much too low
 - For reasonable calibrations of wealth, households “save out of borrowing constraint”
- Kaplan-Violante argue that incorporating illiquid wealth into model is key:
 - Liquid assets (-1.5% real return)
 - Illiquid asset (2.3% real return)
 - Unsecured debt (6% real interest rate)
- Two types of hand-to-mouth agents:
 - Poor hand-to-mouth (no illiquid wealth, no liquid wealth)
 - Wealthy hand-to-mouth (have illiquid wealth, no liquid wealth)
- This model generates much higher rebate responses

Aguiar & Hurst (2013)

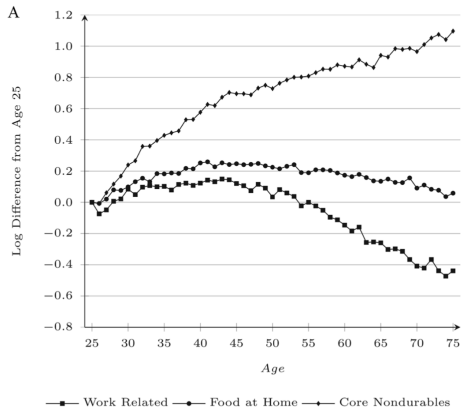


FIG. 2.—Life cycle profiles of three subaggregates. Panels *A* and *B* are identical to panels *A* and *B* of figure 1, respectively, except that we disaggregate nondurable consumption into three categories. The categories are food at home (circles); work-related expenses (squares), which include transportation, food away from home, and clothing/personal care; and core nondurables (diamonds), which include all other categories of total nondurable expenditure (including housing services but excluding alcohol and tobacco). See the caption of figure 1 for additional sample and estimation descriptions.

■ Hump shaped consumption profile mostly driven by at work expenses!

Conclusion

- Models of consumption:
 - Random walk hypothesis
 - Precautionary savings
 - Liquidity constraints
- Evidence:
 - Time series evidence on random walk hypothesis
 - Micro evidence:
 - MPCs out of transitory income
 - Structural estimation using life cycle profile